

기초전기실험

11주차

권혁준 교수님 반 (화요일)
오선우 조교

11 주차 실험 목표

■ AC Ch. 7 : The oscilloscope and phase measurements

- Oscilloscope에 2개의 probe를 연결하여 두 정현파 사이의 위상차를 측정한다.
- Lissajous pattern을 관찰한다.
- R-C 회로에서 저항 값을 변화 시킬 때 위상의 변화를 관찰한다.

■ AC Ch. 8 : Series sinusoidal circuits

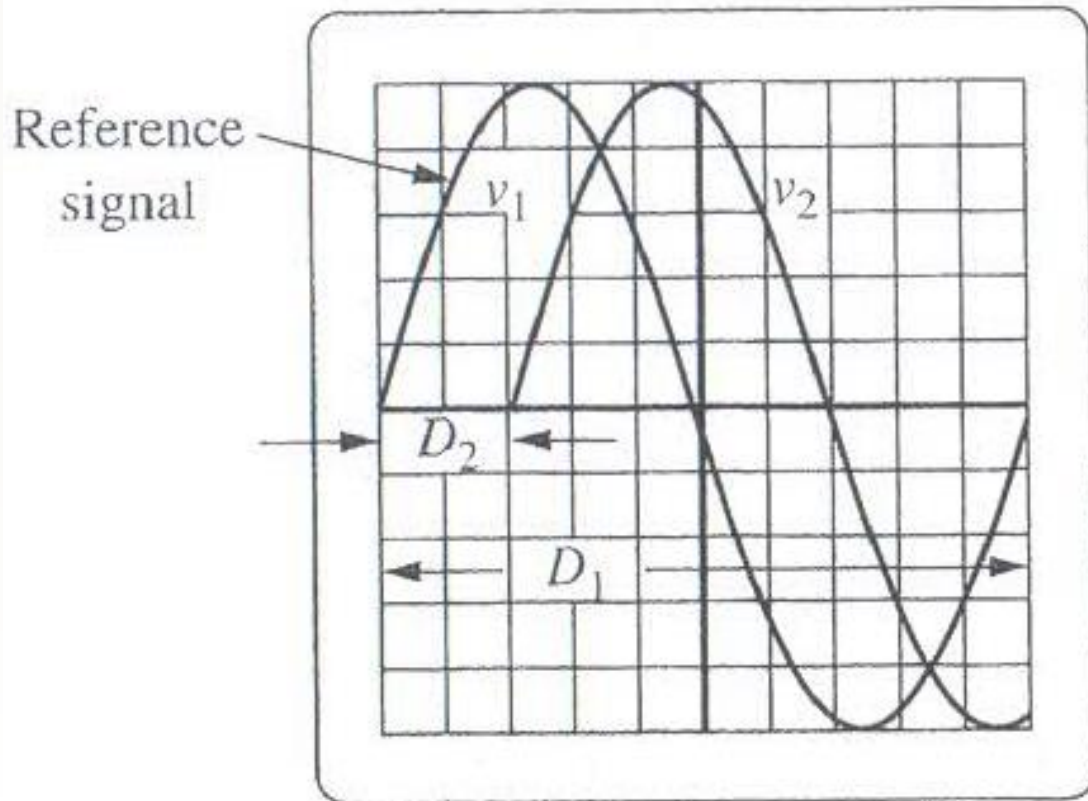
- 직렬로 연결된 R-L, R-C, R-L-C 회로에 교류 전원을 가할 때, Kirchhoff's voltage law가 성립함을 확인한다.
- 각 소자에 걸리는 전압과 회로에 흐르는 전류를 측정하고, 위상차를 구한다.

■ AC Ch. 9 : Parallel sinusoidal circuits

- 병렬로 연결된 R-L, R-C, R-L-C 회로에 교류 전원을 가할 때, Kirchhoff's current law가 성립함을 확인한다.
- 각 소자에 걸리는 전압과 회로에 흐르는 전류를 측정하고, 위상차를 구한다.

PHASE MEASUREMENTS

■ Dual-Trace Method

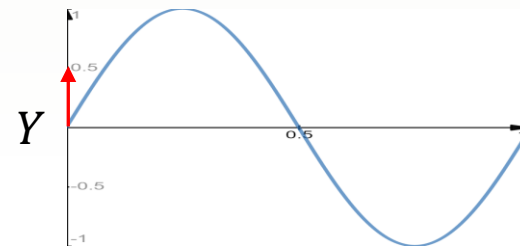
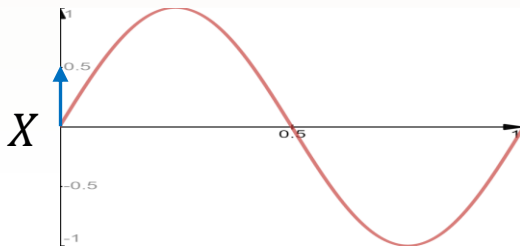
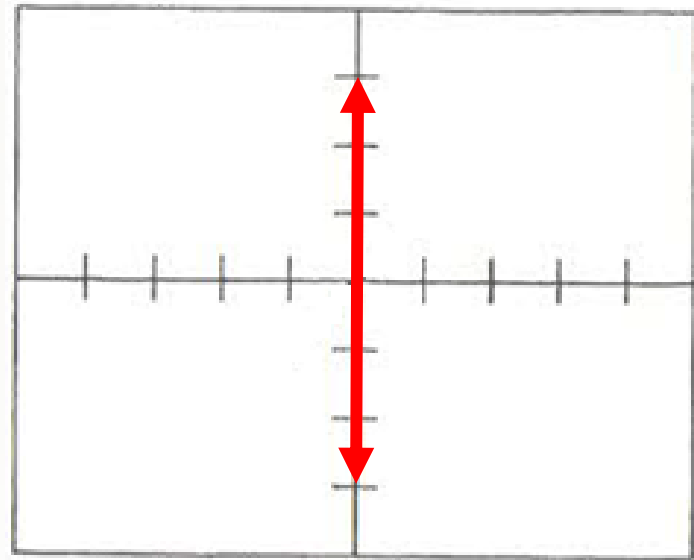
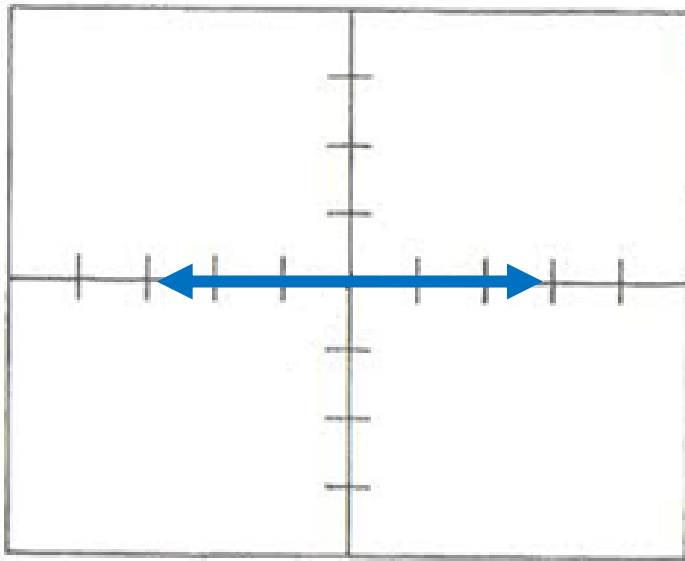


$$D_1 : D_2 = 360^\circ : \theta$$

$$\therefore \theta = \frac{D_2}{D_1} \times 360^\circ$$

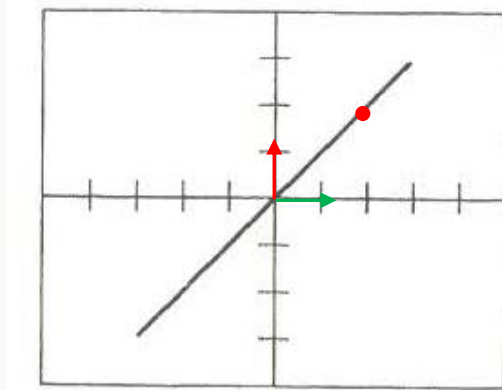
PHASE MEASUREMENTS

▣ Lissajous-Pattern Method

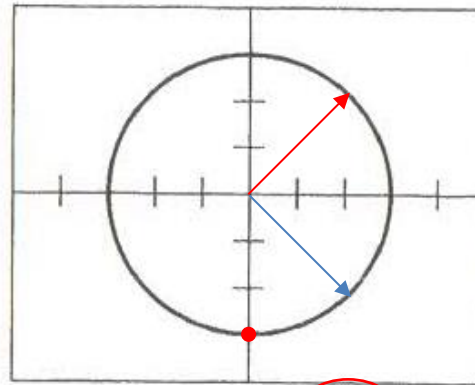


PHASE MEASUREMENTS

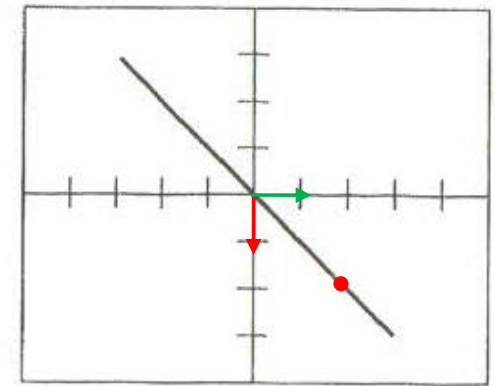
■ Lissajous-Pattern Method



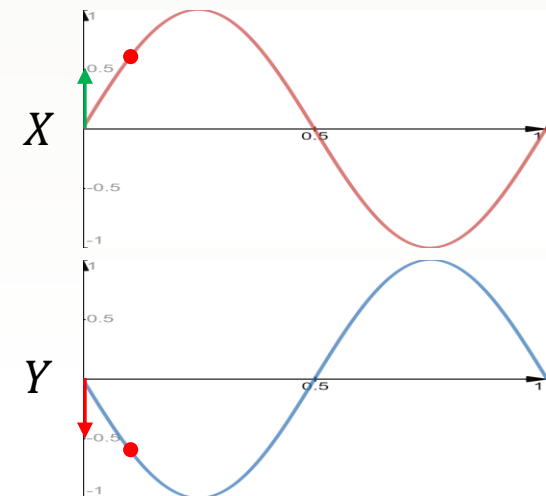
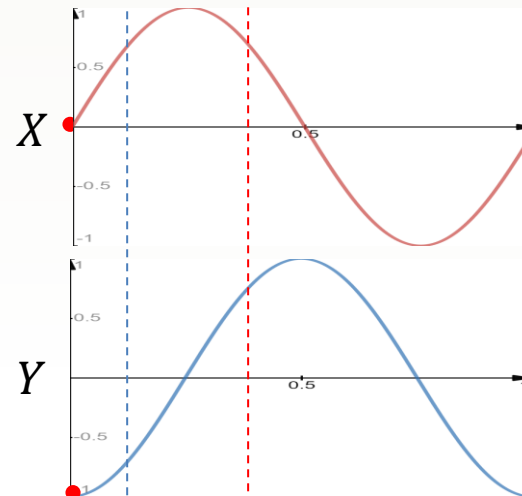
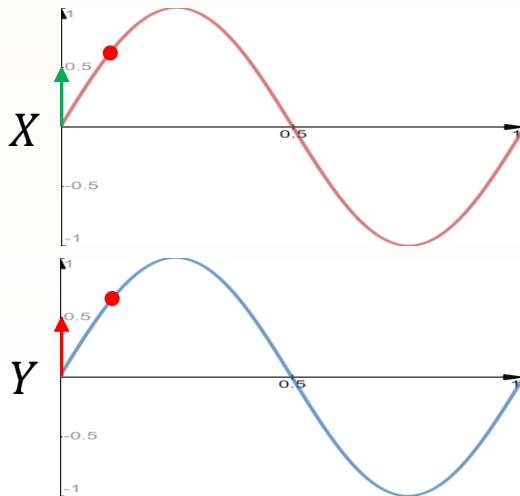
$\theta = 0^\circ \text{ or } 360^\circ$



$\theta = 90^\circ \text{ or } 270^\circ$

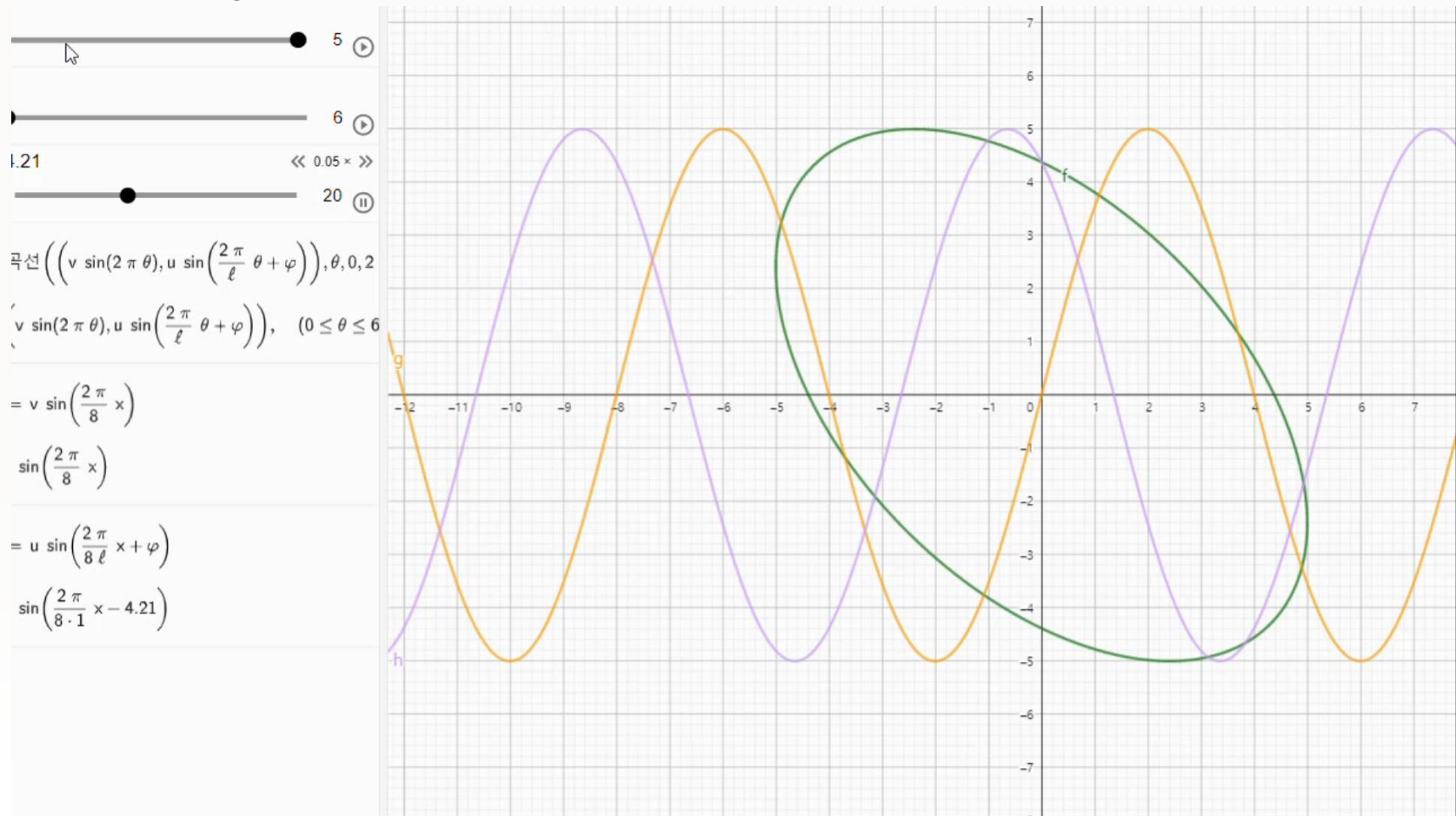


$\theta = 180^\circ$



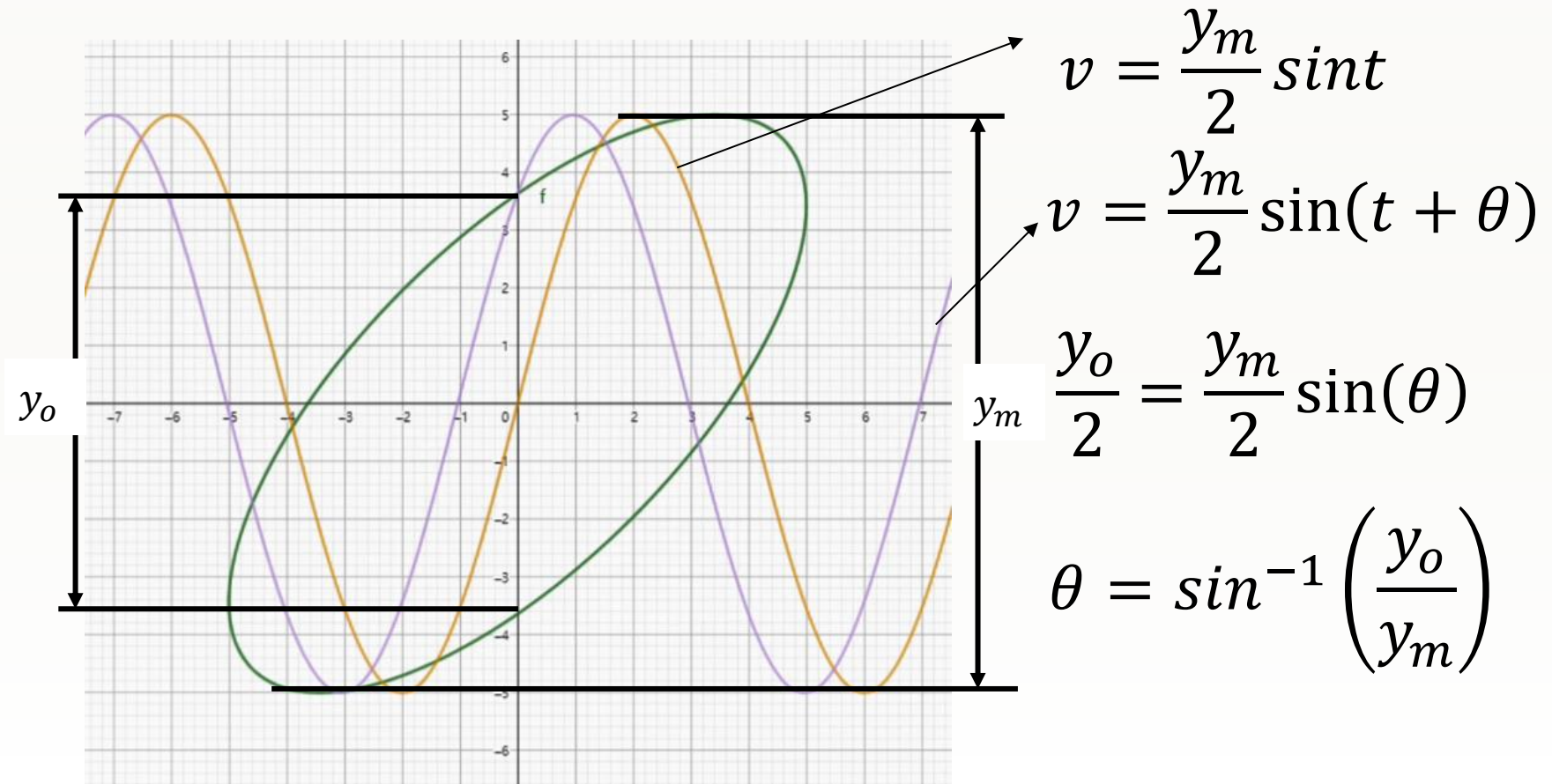
PHASE MEASUREMENTS

▣ Lissajous-Pattern Method



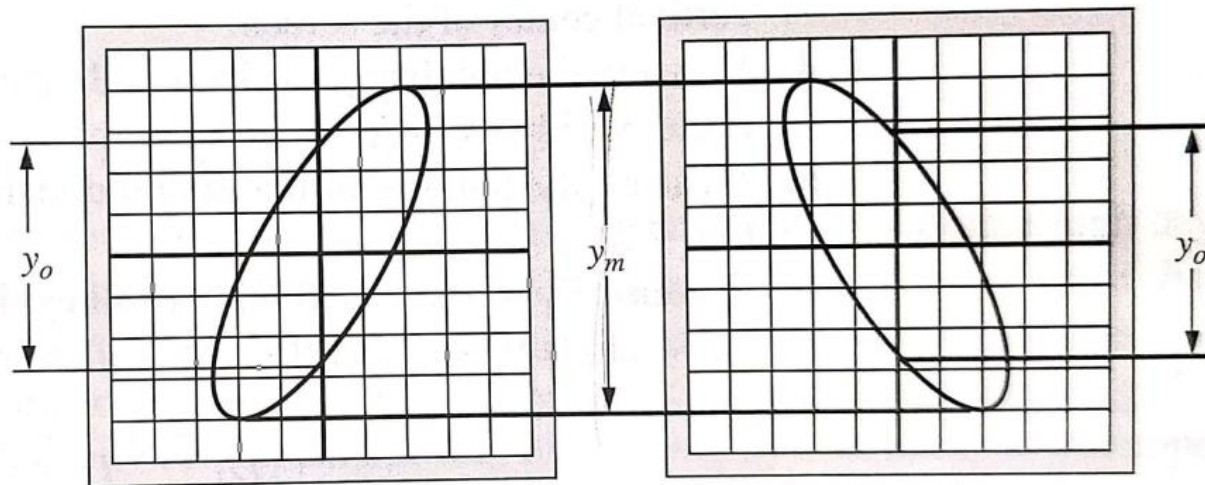
PHASE MEASUREMENTS

■ Lissajous-Pattern Method



PHASE MEASUREMENTS

■ Lissajous-Pattern Method



$0^\circ < \theta < 90^\circ$ or
 $270^\circ < \theta < 360^\circ$

$$\theta = \sin^{-1} \frac{y_o}{y_m} \quad (7.3)$$

$90^\circ < \theta < 180^\circ$ or
 $180^\circ < \theta < 270^\circ$

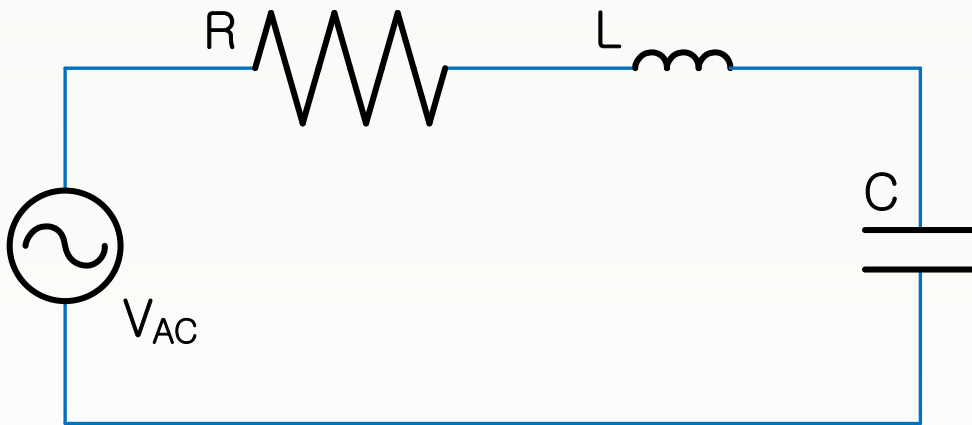
$$\theta = 180^\circ - \left(\sin^{-1} \frac{y_o}{y_m} \right) \quad (7.4)$$

OSCILLOSCOPE



Horizontal에서 Horiz를 활성화
xy 모드로 변경

SERIES SINUSOIDAL CIRCUIT



Voltage

$$E^2 = V_R^2 + (V_L - V_C)^2$$

$$E = \sqrt{V_R^2 + (V_L - V_C)^2}$$

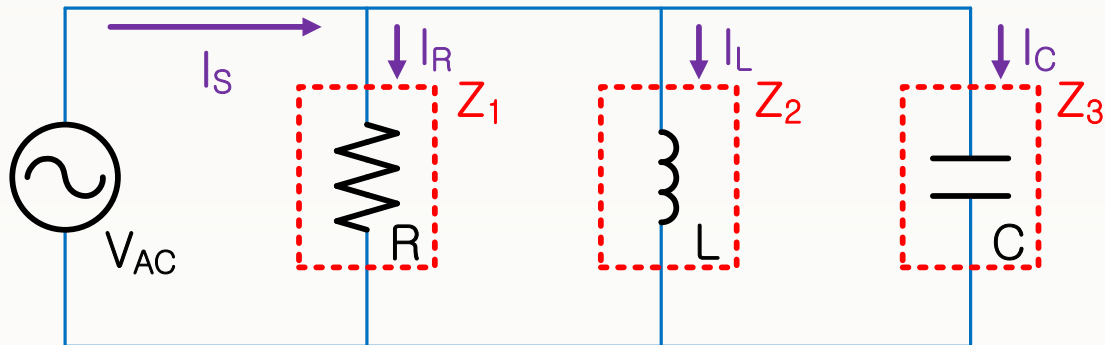
Impedance

$$Z^2 = R^2 + X_T^2$$

Reactance

$$X_T = X_L - X_C$$

PARALLEL SINUSOIDAL CIRCUIT



Impedance

$$\frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3}$$

Current

$$I_S = \sqrt{I_R^2 + (I_L - I_C)^2}$$

11 주차 실험

■ AC Ch. 7 : The oscilloscope and phase measurements

- part1($1\text{k}\Omega \times 1$, $3.3\text{k}\Omega \times 1$, $6.8\text{k}\Omega \times 1$, $0.47\mu\text{F} \times 1$)
 - FIG 7.9 구성 후 $E_S = 4V_{p-p}$ (200Hz) 인가
 - 저항별 전압, 위상차 측정 후 Table 7.1, 7.2, 7.3, Graph 7.1 완성
- part2($1\text{k}\Omega \times 1$, $3.3\text{k}\Omega \times 1$, $6.8\text{k}\Omega \times 1$, $0.47\mu\text{F} \times 1$)
 - FIG 7.11 구성 후 $E_S = 4V_{p-p}$ (200Hz) 인가
 - 저항별 전압, 위상차 측정 후 Table 7.4, 7.5, 7.6 완성
- part3($1\text{k}\Omega \times 1$, $3.3\text{k}\Omega \times 1$, $6.8\text{k}\Omega \times 1$, $0.47\mu\text{F} \times 1$)
 - FIG 7.11 구성 후 $E_S = 4V_{p-p}$ (200Hz) 인가
 - Lissajous-Pattern Method를 통해 위상차를 측정하여 Table 7.8 완성

■ AC Ch. 8 : Series sinusoidal circuits

- part1($1\text{k}\Omega \times 1$, $10\text{mH} \times 1$)
 - FIG 8.1 구성 후 $E_S = 4V_{p-p}$ (10kHz) 인가
 - 측정을 통해 Table 8.1 완성
- part2($1\text{k}\Omega \times 1$, $0.01\mu\text{F} \times 1$)
 - FIG 8.2 구성 후 $E_S = 4V_{p-p}$ (10kHz) 인가
 - 측정을 통해 Table 8.3 완성
- part3($1\text{k}\Omega \times 1$, $10\text{mH} \times 1$, $0.01\mu\text{F} \times 1$)
 - FIG 8.3 구성 후 $E_S = 4V_{p-p}$ (10kHz) 인가
 - 측정을 통해 Table 8.5 완성

■ AC Ch. 9 : Parallel sinusoidal circuits

- part1($1\text{k}\Omega \times 1$, $10\Omega \times 1$, $10\text{mH} \times 1$)
 - FIG 9.1, 9.2, 9.3 구성 후 $E_S = 4V_{p-p}$ (10kHz) 인가
 - 측정을 통해 Table 9.1, 9.2, 9.4 완성